

ITRANOX 10MG/ML SOLUTION

Description and Composition

ITRANOX 10MG/ML SOLUTION is a yellow to amber clear solution. Each ml contains 10mg Itraconazole as its active ingredient.

Pharmacodynamics

The mode of action of itraconazole is based on its highly selective binding ability to fungal Cytochrome P-450 iso-enzymes. This inhibits the synthesis of ergosterol and affects membrane-bound enzyme function and membrane permeability. This effect is irreversible and causes structural degeneration.

Pharmacokinetics

Laboratory animals rapidly absorb orally administered itraconazole. It binds very extensively to plasma proteins (>99 %) and distributes to tissues. More than 30 metabolites are formed, from which hydroxyitraconazole has an antifungal activity as the parent. Excretion is rapid and mainly via the faeces.

In cats a single oral dose of 5 mg/kg results in maximum plasma concentrations of on average 0.525 mg/l attained 2 hours after dosing. The AUC_{0-24 h} is 5 mg.h/l. The half-life in plasma is about 12 hours. After repeated administration for one week at 5 mg/kg/day, the maximum plasma concentration is more than doubled. The AUC_{0-24 h} is increased 3 times to 15 mg.h/l and the plasma half-life is also increased 3 times to 36 hours.

Indication

For the treatment of dermatophytosis caused by *Microsporum canis* in cats.

Recommended Dose

The solution is administered orally directly into the mouth by means of a dosing syringe. The daily dosage is 5mg/kg or 0.5ml/kg/day. The dosage regime is 0.5 ml/kg/day for 3 alternate periods of 7 consecutive days, each time with 7 days without treatment in between.

7 days	7 days	7 days	7 days	7 days
Treatment	No treatment	Treatment	No treatment	Treatment

The dosing syringe shows graduations per 100 gram of body weight. Fill the syringe by pulling the plunger until it reaches the graduation corresponding to the correct body weight of the cat. When administering the product to kittens, the administrator should be careful not to administer more than the recommended dose/weight. For kittens weighing less than 0.5kg, a 1ml syringe which allows proper dosing should be used.

Treat the animal by slowly and gently injecting the liquid into the mouth, allowing the cat to swallow the product. After dosing, the syringe should be removed from the bottle, washed and dried and the cap should be screwed back on tightly.

Data in humans shows that food intake may result in lower drug absorption. Therefore, it is recommended to administer the product by preference between meals.

In some cases, a prolonged time between clinical and mycological cure may be observed. In cases where a positive culture is obtained 4 weeks after the end of administration, the treatment should be repeated once at the same dosage regimen. In such cases where the cat is also immune-suppressed, treatment should be repeated and the underlying disease addressed.

Route of Administration

For oral administration.

Contraindications

Do not administer to cats with hypersensitivity to itraconazole or one of the other ingredients.

Do not administer to cats with impaired liver or kidney function.

For use in pregnant and lactating queens: see section 'Pregnancy and Lactation'.

Warnings and Precautions**Special warnings for each target species:**

Some cases of feline dermatophytosis can be difficult to cure, especially in catteries.

Cats treated with itraconazole can still infect other cats with *M. canis* as long as they are not mycologically cured. It is therefore advised to minimise the risk of re-infection or spread of infection by keeping healthy animals (including dogs as they can also be infected by *M. canis*) separate from cats that are being treated. Cleaning and disinfection of the environment with appropriate fungicidal products is highly recommended – especially in case of group problems.

When clipping the hair of infected cats, the advice of the veterinarian should be sought first.

Clipping of the hair coat is considered useful because it removes infected hairs, stimulates new hair growth and hastens recovery. It is strongly recommended that clipping is performed by a veterinarian. In cases with limited lesions, hair clipping can be limited to the lesions only, whereas in cats with generalized dermatophytosis it is recommended to clip the entire hair coat. Care should be taken not to cause trauma to the underlying skin during clipping. It is recommended that disposable, protective clothing and gloves are worn during the clipping of the affected animals. The clipping of the hair should be performed in a well-ventilated room which can be disinfected after clipping. The hairs should be disposed of appropriately and all instruments, clippers etc. should be disinfected.

Treatment of dermatophytosis should not be limited to treatment of the infected animal(s). It should also include disinfection of the environment with appropriate fungicidal products, since *M. canis* spores can survive in the environment for up to 18 months. Other measures such as frequent vacuuming, disinfection of grooming equipment and removal of all potentially contaminated material that cannot be disinfected will minimize the risk of re-infection or spread of infection. Disinfection and vacuuming should be continued for an extended period after the cat is clinically cured, but vacuuming should be limited to surfaces, which may not be cleaned with a damp cloth. All other surfaces should be cleaned with a damp cloth. Any cloth used for cleaning should be washed and disinfected or disposed of and the used vacuum cleaner bag should be disposed of.

Measures to prevent introduction of *M. canis* into groups of cats may include isolation of new cats, isolation of cats returning from shows or breeding, exclusion of visitors and periodic monitoring by Wood's lamp or by culturing for *M. canis*.

In refractory cases the possibility of an underlying disease should be considered.

Frequent and repeated use of an antimycotic may result in the induction of resistance to antimycotics of the same class.

Refer to section 'Special precautions to be taken by the person administering the veterinary medicinal product to animals' for Operator warnings.

Special precautions for use**Special precautions for use in animals:**

Cats suffering from dermatophytosis, but also in poor general condition and/or suffering from additional diseases or impaired immunological response should be monitored closely during treatment. Because of their condition, this category of animals may be more sensitive to the development of adverse effects. In

case of a serious adverse effect, treatment should be interrupted and supportive care therapy (fluid therapy) should be initiated if necessary. If clinical signs suggestive of liver dysfunction develop, treatment should be discontinued immediately. It is very important to monitor liver enzymes in animals showing signs of liver dysfunction.

Cats suffering from heart diseases should be carefully monitored and the treatment should be withdrawn if the clinical signs deteriorate.

Special precautions to be taken by the person administering the veterinary medicinal product to animals: If a suspected lesion occurs on a human, consult a physician, since *M. canis* dermatophytosis is a zoonotic disease. Therefore, wear latex gloves when clipping hair of infected cats, when handling the animal during treatment or when cleaning the syringe.

Wash hands and exposed skin after use. In case of accidental contact with eyes, rinse thoroughly with water. In case of pain or irritation, seek medical advice. In case of accidental ingestion, rinse mouth with water.

Interactions with Other Medicaments

Vomiting, hepatic and renal disorders were seen after concomitant treatment of Itraconazole and cefovecin. Symptoms like motor incoordination, faecal retention and dehydration are observed when tolfenamic acid and Itraconazole are given simultaneously. Co-administration of the product and these drugs, in absence of data in cats, should be avoided.

Pregnancy and Lactation

Do not use in pregnant or lactating queens. Malformations and foetal resorptions were seen in overdose studies in laboratory animals.

Side Effects

In clinical studies, some form of adverse reaction possibly related to the administration of the product were noted. Common adverse reactions were vomiting, diarrhoea, anorexia, salivation, depression and apathy. These effects are usually mild and transient. In very rare cases, a transient increase in liver enzymes may occur. In very rare cases, this was associated with icterus. If clinical signs suggestive of liver dysfunction develop, treatment should be discontinued immediately.

Symptoms and Treatment of Overdose

After a 5x overdose of itraconazole administered for 6 consecutive weeks, reversible clinical side effects were: rough hair coat, decreased food intake and reduced body weight. A 3x overdose for 6 weeks did not result in clinical side effects. Both after a 3x and a 5x overdose for 6 weeks, reversible change in serum biochemical parameters indicating liver involvement occur (increased ALT, ALP, bilirubin and AST). At 5 times overdose, a slight increase in segmented neutrophils and a slight decrease in lymphocytes were observed.

No studies on overdose in kittens have been performed.

Storage Condition

Store below 30°C. Protect from light.

Shelf life

3 years.

Discard 5 weeks after first opening.

Packing

60ml, 100ml, 200ml, 500ml, 1L.

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